

Matee ur Rasool Meladi

Electronic Engineering Undergraduate Student
The Islamia University of Bahawalpur (IUB), Pakistan

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Academic Background

[Google Scholar](#)

• BSc in Electronic Engineering

The Islamia University of Bahawalpur (IUB), Pakistan

2023 – 2027 (Expected)

CGPA: 3.83/4.00 (6 semesters)

Research Experience

• CSI-Based Human Activity Recognition (CSI-HAR) Research Project — Group Team Leader | Communication Lab, IUB

Mar 2026 – Aug 2026

Supervised by: Dr. Abbas Abbasi, Assistant Professor, Department of Electronic Engineering, IUB.

- Led CSI-based HAR research, covering data preprocessing, ML/DL, and Raspberry Pi real-time implementation.
- Technology Used: Raspberry Pi 4, WiFi CSI, Python, ML, DL, Signal Processing, iperf3.

• PSO-Based Battery RUL Prediction – Published Research Paper, Vehicle

6 Aug 2026

Supervised by: Dr. Zeeshan Ahmad Arfeen, Department of Electrical Engineering, IUB, Pakistan

- Developed a PSO-based machine learning framework for lithium-ion battery RUL prediction.
- Technology Used: Python, ML, PSO, Feature Engineering, and Battery Data Analysis.

Industrial Experience

• Engineering Intern, Pakistan Aeronautical Complex (PAC), Kamra

15 Jun 2026 – 24 Jul 2026

- Studied avionics, radar systems, PCB testing, electronic fault diagnosis, and aircraft communication interfaces.

• Training: Biomedical Instrumentation & PID Control — NUST | 2 Weeks

10 Aug 2026 – 20 Aug 2026

- Applied signal processing, PID control, feature extraction, and AI integration.

Academic Projects

• FPGA Implementation of a 3-Bit Up/Down Counter – Digital System Design

2026

- Implemented a synchronous 3-bit up/down counter on FPGA with LED and seven-segment outputs.
- Technology Used: Verilog HDL, Xilinx ISE 14.7, Spartan-3 FPGA.

• Segway Self-Balancing System – Control Systems

2026

- Modelled a Segway inverted pendulum and designed a feedback stability controller.
- Technology Used: MATLAB/Simulink, Control System Design, Feedback Control.

• Feedback-Controlled Buck Converter Design – Power Electronics

2026

- Designed a feedback-controlled DC-DC buck converter for 50 V–15 V regulated output.
- Technology Used: MATLAB, pspice software, SG3525/UC3845, Feedback Control, Power Electronics.

• Signal Analyzer Application – MATLAB App Designer

2025

- Developed a signal analysis app with FFT, energy/power analysis, and classification.
- Technology Used: MATLAB, MATLAB App Designer, FFT, Signal Processing.

• PLC–ESP32 Integration for Industrial Automation Workcell – CARE Lab

2025

- Integrated PLC-based workcell with ESP32 for industrial automation and IoT communication.
- Technology Used: PLC, ESP32, Industrial Automation, IoT.

• Full-Stack School Management System

2025

- Developed a web-based student management system with authentication and RESTful APIs.
- Technology Used: React, Python, FastAPI, SQLAlchemy, REST API, JWT.

• AI Smart Environment and Mood Automation System – Embedded Systems

2025

- Developed an AI-enabled environmental monitoring system with anomaly detection and adaptive learning.
- Technology Used: ESP32 DevKit V1, PIR Sensor, MQ-2 Sensor, Z-Score Detection, EMA, ThingSpeak, Wi-Fi.

• Reminder Application for Time Management – Software Engineering

2024

- Developed a To-Do and reminder app with authentication, scheduling, and notifications.
- Technology Used: C++, Qt, GUI Development, Database Management.

Technical Skills and Areas of Expertise

Languages: Urdu (Native), English (Working Proficiency).

Programming Languages: Python, C++, OOPs, Assembly Language, MATLAB.

Courses & Workshops: Signal Processing Onramp (MATLAB), Siemens NX: Engineering Essentials & Part Design (Siemens), Remote Sensing & Disaster Management with Python (IEEE GRSS Lahore Chapter), Engineering and Product Design Processes (Arizona State University), AI for Everyone (DeepLearning.AI), Project Management (Rice University).

Areas of Competence: Power Electronics, Control Systems, Machine Learning, AI / Signal Processing, IoT Integration, Sensor Data Acquisition, Embedded Systems, Circuit Design.

Soft Skill: Problem-solving, Self-learning, Presentation, Research, Openness to interdisciplinary learning, Teamwork.

Activities and Achievements

- **Honhaar Scholarship** – Recipient of the merit-based scholarship in recognition of academic performance.
- **Prime Minister's Laptop Scheme** – Selected among 100,000 meritorious students nationwide on 100% merit.
- **Research Publication** – Published research on Lithium-Ion Battery Remaining Useful Life (RUL) prediction using Particle Swarm Optimization and hybrid Machine Learning approaches.
- **Group Team Leader, CSI-Based Human Activity Recognition** – Led team coordination, data collection, signal processing, and Machine/Deep Learning experimentation.